Health-a-thon 2026

FINAL SUBMISSION

RAD-SMART

Radiotherapy appointments that hold: one daily time per patient, a new plan in seconds when the machine breaks down, and messages in the patient's own language

TEAM NAME RAD-SMART

TEAM CODE YU8PAK

Team Overview & Background

Doctors from the department RAD-SMART is built for, and the engineers who built it with them.

Dr Akshay Dinesan Team leader, doctor partner	Wrote the problem statement from daily work; owns the clinical rules and the pilot
Dr Shirley Lewis Salins Doctor partner	Clinical review of the department's rules and the design
Dr Umesh Velu Doctor partner	Clinical review of the department's rules and the design
Abhinand T M Technical lead	Coordinates the team, the doctors and the submissions; reviews the build
P S Raghotham Rao Lead engineer	Built the scheduling engine, the digital twin, the CARE plugin and the Sarvam AI layer

The department

Radiotherapy, Kasturba Medical College, Manipal (MAHE). One Elekta Versa HD treats 70 to 90 patients a day, 08:30 until about 01:00; a second machine arrives in 2027.

The doctor partner's role

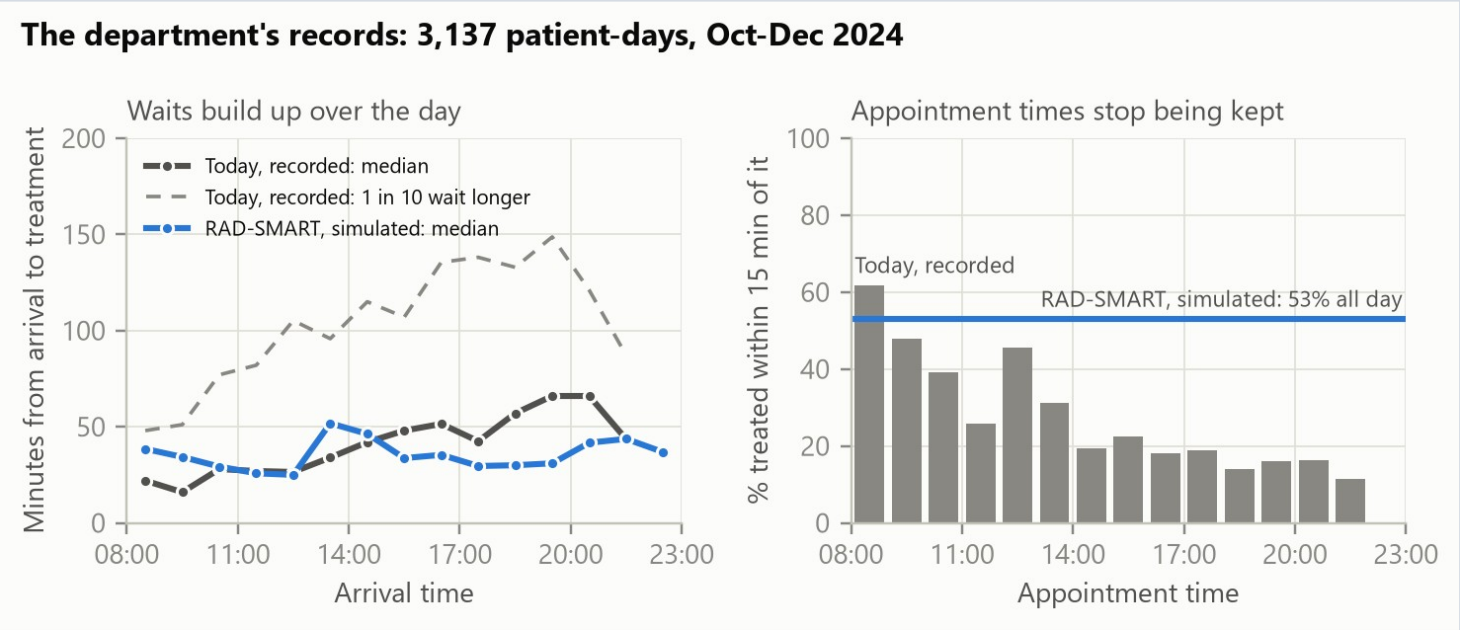
Dr Akshay Dinesan brought the problem and the department's own records. The technologists' and doctors' written answers (21 Sep, 24 Oct) are in the code as rules, and their feedback (27 Oct) added preferred times and treating patients out of turn.

The platform

CARE, the open-source hospital information system of the Open Healthcare Network. RAD-SMART is a plugin: it uses CARE's patients, staff and roles, with no change to CARE.

Problem Statement and Supporting Evidence

One machine, 70 to 90 patients a day, and the delay builds up every day of a one to seven week course.



55 working days, used only as anonymised totals.

109 min

one patient in ten waits longer than this, from arrival to treatment

22%

are treated more than an hour after their appointment time

69%

of India's radiotherapy facilities have only one machine, like this one

Times are given by hand, without knowing how long a session takes. A breakdown is handled by phone, and patients already on the bus are not told.

Target Users and Use Case

Five people shape a patient's day. Each gets one screen inside CARE, which the hospital already uses.

Who	What they need	What RAD-SMART gives them
Radiation oncologist	A time that fits the treatment plan	Enters the course from the patient's visit; approves a first day and one daily time
Technologist	To run the machine's day, and a fast answer when it breaks	The day board: Call next, Priority for an emergency, a re-plan in seconds after a breakdown
Reception	Fast, certain check-in	Scans the code on the patient's phone
Patient and family	To know when to come, in their language	One private link: times, voice notes, one-tap replies, morning, afternoon or evening
Head of department	Evidence that long waits are falling	Insights from the times the board records

Setting: a radiotherapy department with one linear accelerator, every weekday from 08:30 to about 01:00, running CARE.

Solution Overview, Key Features & Impact

RAD-SMART predicts, plans, re-plans and tells. A person approves every change.

1 Predict

how many minutes each session will take, from the technique, site and the patient's needs.

2 Plan

each course a first day and one daily time, with an optimiser that cannot break a department rule.

3 Re-plan

the rest of the day in about 10 to 14 seconds when the machine breaks down.

4 Tell

patients in Kannada, Malayalam, Hindi or Konkani, with a voice note, on a phone page and WhatsApp.

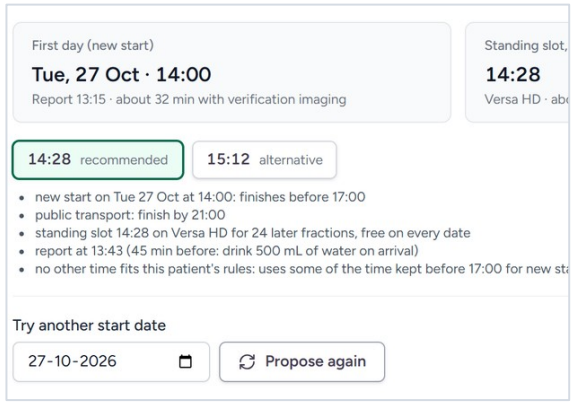


The day board on the live demonstration: who is in the room, whom to call next, and the rule check.

RAD-SMART decides only **when** a patient is treated, never whether or how. Staff can always override it: take a patient in out of turn, or put one first.

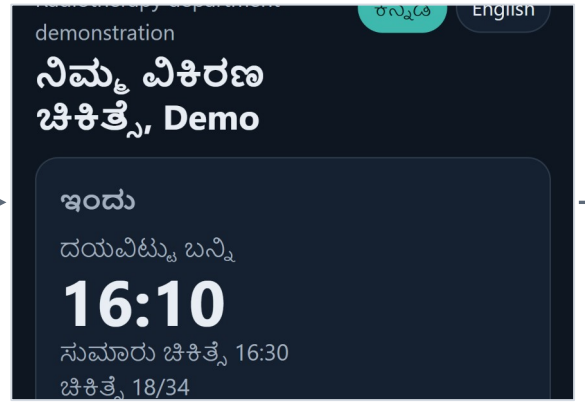
Solution Overview, Key Features & Impact

From the doctor's prescription to the patient's phone, and what it does to the waiting.



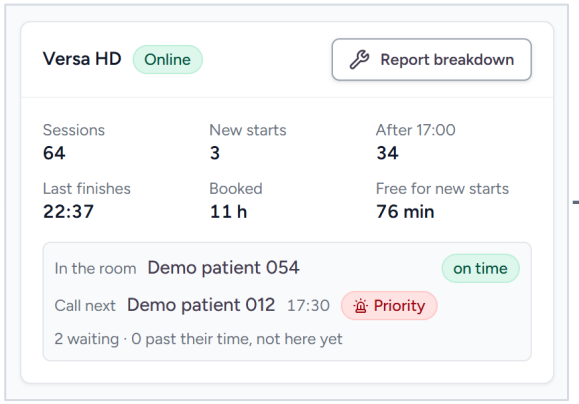
Doctor

Enters the course; approves a first day and a daily time, with reasons



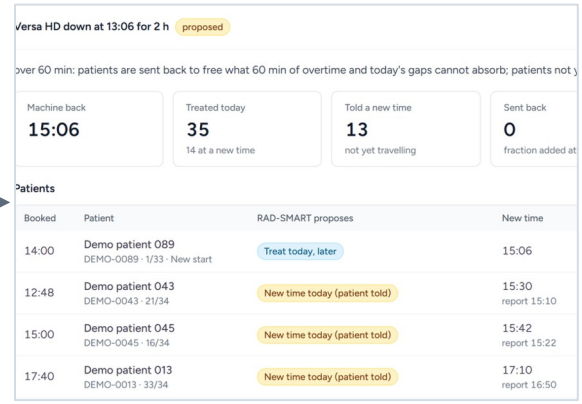
Patient

One private link: the time to come, a voice note, a QR code



Technologist

Call next, out of turn with a reason, Priority for an emergency



Breakdown

A re-plan under the department's rules; approved by a person

−35%
90th-percentile wait, 108.6 to 70.4 min, on the department's own 55 days

22% to 5.7%
of patients treated more than an hour late

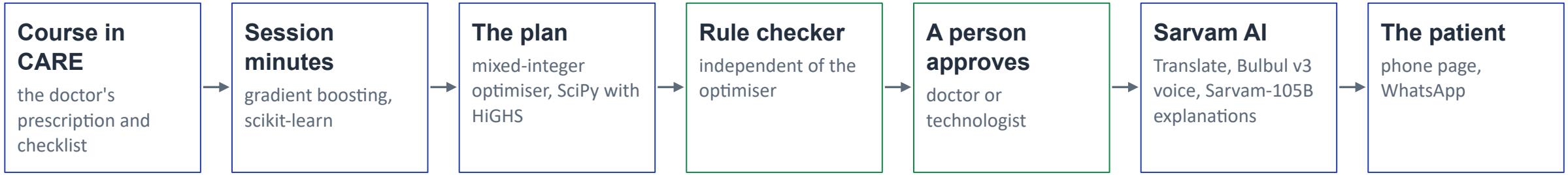
29% to 50%
treated within 15 minutes of their time

104 to 39 min
median wait after a 150-minute breakdown, with a new plan in 14 s

Expected impact: the department's recorded days re-planned in a digital twin calibrated on them, and synthetic breakdown days. A pilot measures it live.

Technology, Tools and AI Approach

Each part does what it is best at: the optimiser guarantees the rules; language models only translate and explain.



Green: the safety gate. A plan that breaks a hard rule cannot be produced, and nothing reaches a patient before a person approves it.

Why this mix

- Only optimisation guarantees the rules; only language models talk to patients in their language. We scored five approaches and this hybrid was the only one that does both.
- The language models never schedule. Every translation and explanation is checked for changed times.

Data

- The department's records: 55 days, anonymised totals, to calibrate the digital twin.
- Synthetic patients for training and every demonstration.
- Real session minutes, recorded by the board's In room and Out taps, for retraining in the pilot.

Stack and integrations

- Django and React plugins in CARE; PostgreSQL.
- One Docker image on Microsoft Azure Container Apps.
- WhatsApp from the department's phone; QR check-in; Telegram in the demonstration only.

Build Sprint Journey and Progress

From a Round 1 engine to a working app inside CARE, live on Azure and shaped by the department.

19 Sep	5 Oct	24 Oct	25 Oct	26 Oct	27 Oct	3 Nov
Round 1: engine, digital twin	Shortlisted	Department's rules; app inside CARE	Sarvam AI, patient page, QR check-in	Live on Microsoft Azure	Preferred times, live queue	Priority override, emergency starts, guided demo

Working now

- Course intake with a standing daily time, approved by a doctor
- Day board: Call next, out of turn, Priority, rule check
- Breakdown re-planning under the department's rules
- Preferred part of the day, with better-time offers
- Messages in four Indian languages, with voice notes
- Patient phone page, QR check-in, Insights

How it changed

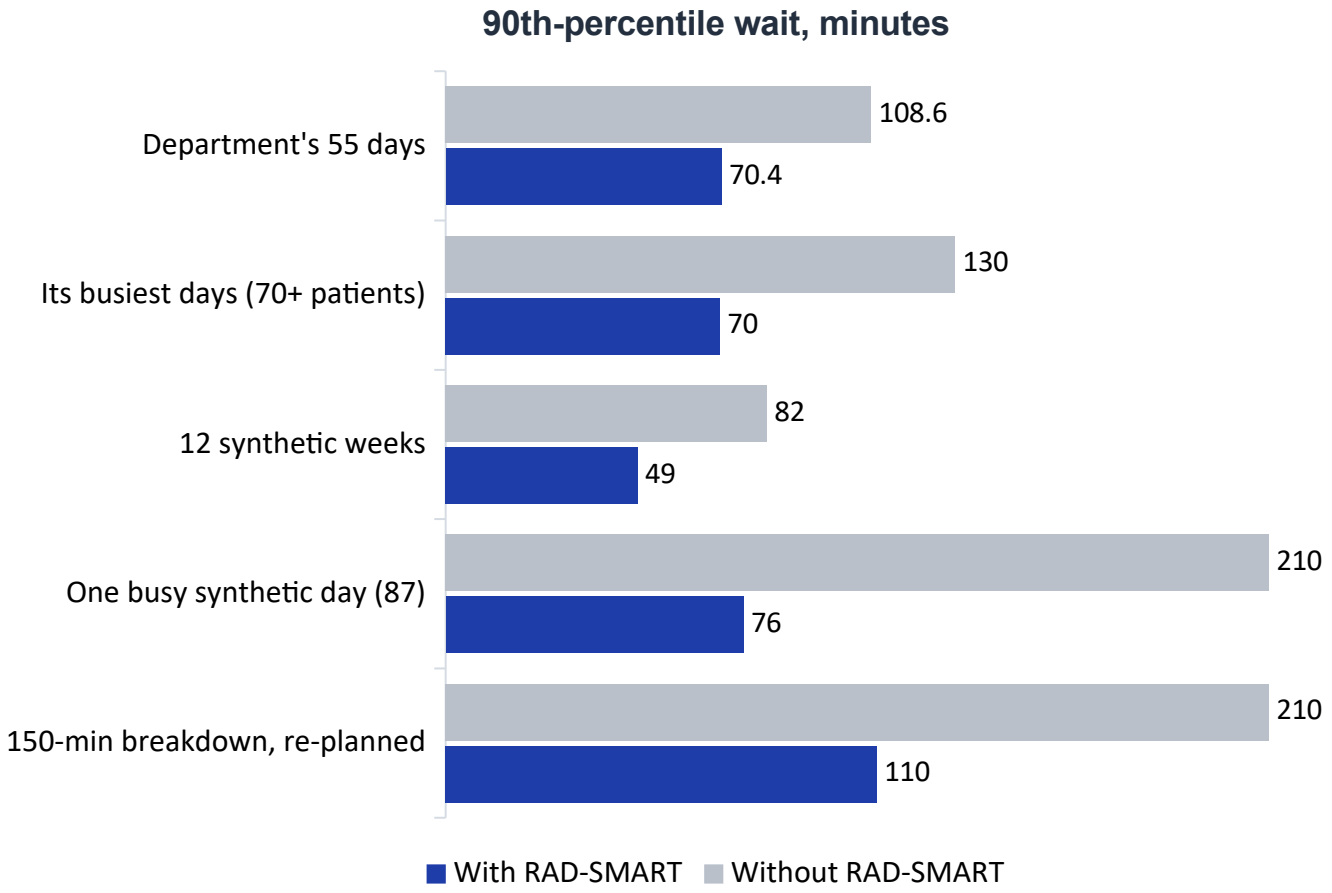
- Round 1 was an engine and a simulation; now it runs inside CARE
- The department's answers became rules: one standing time, new starts by 17:00, the 60-minute breakdown rule
- Testing fixed four translation and wording problems before anyone saw them

Not built yet

- Automatic WhatsApp sending (needs Meta verification)
- Spoken replies from patients
- Reading the oncology system (MOSAIQ, ARIA)
- Live planning for two machines

Evaluation and User Validation

Replayed on the department's own days and on synthetic ones: long waits fall, and no rule is broken.



−35%

90th-percentile wait on the department's 55 days, 108.6 to 70.4 min

14 s

for a new plan after a breakdown; median wait 104 to 39 min

0

hard rules broken on any evaluation day, by an independent checker

Method: the same patients and events with and without RAD-SMART, in a digital twin that reproduces today's long wait (106 against 109 min).

Evaluation and User Validation

The AI parts are measured, and the department's feedback changed the design.

Part	Test	Result
Session-minute prediction	2,000 held-out synthetic sessions	2.15 min mean error, against 2.62 for a table of averages
Optimiser	87-patient day, about 4,800 variables	Proven optimum in 7.1 s on a laptop
Sarvam Translate	112 messages in Kannada, Malayalam, Hindi and Konkani	112 of 112 kept every time; back-translation chrF 76 to 80
Sarvam-105B explanations	20 synthetic plans	20 of 20 accepted, 0 invented times
Safety guards	49 automated tests, the engine's run by GitHub on every change	All pass

Limits, said plainly

- A replay, not a trial: the pilot measures the gain live.
- Session lengths are synthetic until the board has recorded real ones.
- Still to come: native speakers rating every translation, and a technologist trial.

What the department changed

- **21 Sep and 24 Oct**, written answers from the technologists and doctors: one standing time per course, new starts by 17:00, the 60-minute breakdown rule, who approves what.
- **27 Oct**, the doctor partner: patients choose morning, afternoon or evening; a late patient's turn goes to whoever is ready.
- **3 Nov**: staff can put a patient first, and an emergency starts the same day.

Fixed in testing

- Kannada suffixes on times; Konkani's “next week” (messages now name the date); an explanation using the proposed time; an unclear button.

Safety and Responsible AI

RAD-SMART proposes and people decide. The language models are fenced in, and patients' details stay out of them.

People decide

- A doctor approves each course and its time; a technologist approves every change on the day
- Hard rules are optimiser constraints, re-checked by an independent checker
- Staff can always override: out of turn with a reason, or Priority; it warns, never blocks
- If a rule breaks in practice, the department returns to manual scheduling

The AI, fenced in

- Language models never schedule; they translate and explain
- A translation that drops or changes a time is discarded: the patient gets English
- An explanation that mentions a time not in the plan is rejected
- Without Sarvam, everything still works in English

Privacy and consent

- No names, hospital or phone numbers go to Sarvam: only the message text
- The patient page shows a first name only, behind a long random link
- Synthetic data in every demonstration; the department's records only as totals
- The pilot follows the DPDP Act 2023 and Rules 2025, with ethics review

Fairness and access

- Older patients, public transport, the hospice bus and time windows are explicit rules
- Messages in Kannada, Malayalam, Hindi and Konkani, with voice notes for those who read little
- Patients choose morning, afternoon or evening; staff agree any change with them

Deployment, Cost and Sustainability Plan

Live today on Microsoft Azure as one Docker image; the pilot runs the same image with demonstration mode off.

To pilot it

- A CARE instance: the hospital's own, or our image; RAD-SMART installs as a plugin
- A laptop or tablet at the console and at reception; a USB scanner or camera
- The department's WhatsApp on a staff phone; internet for Sarvam
- Approval by the head of department, hospital IT and the ethics committee

Integration

- Uses CARE's own patients, users and roles; no change to CARE
- Next: a read-only feed from the oncology system (MOSAIQ or ARIA, HL7 v2 or FHIR), so courses are not typed twice

Scale and sustain

- Versa HD pilot, then the second machine (2027), then other CARE hospitals: 69% of Indian centres have one machine
- Open source, no licence fee; the department's own staff can run it

One-time

Server, or an existing hospital VM	**₹0 to 1,50,000**
Tablets or existing PCs; QR scanner	**₹5,000 to 75,000**
Software licence	**₹0**

Ongoing

Per message: translation and voice note	**about ₹1**
Per course: about 3 messages and explanations	**₹3 to 5**
Per month, on the hospital's server	**about ₹500 to 900**
Per month, on Azure Central India instead	**about ₹4,500 to 7,000**

Assumptions: Sarvam list prices (Translate ₹20 and Bulbul v3 ₹30 per 10,000 characters); a message of about 190 characters; about 1,000 courses a year. Azure: one always-on container (0.5 vCPU) and a small managed database. WhatsApp Business API costs come later.

What Next? Plans to take initiative forward (if any)

Validate with the department, then a 90-day pilot on the Versa HD.

Nov to Dec 2026	Technologist trial and native-speaker ratings; approvals from the head of department, hospital IT and the ethics committee; patient notice and consent for messages
Jan 2027	Pilot, phase 1: two weeks measuring today's waits with the day board
Feb 2027	Phase 2: three weeks in shadow mode, RAD-SMART proposing while staff schedule by hand
Feb to Apr 2027	Phase 3: eight weeks live. Target: a 90th-percentile wait at least 25% below 109 minutes
Later in 2027	WhatsApp Business API, spoken replies, the second machine, an oncology-system feed, and the plugin offered to other CARE hospitals

Make all the solution information publicly available?

Yes

The code, website, deck, research paper and demo video are already public and open source.

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Thank you

RAD-SMART • YU8PAK

Contact: akshay.dinesan@manipal.edu

Try the live demo: https://radsmart.braveglacier-5f9807f7.uaenorth.azurecontainerapps.io/api/care_radsmart/try/

Demo video: https://psraghothamrao8.github.io/RAD-SMART/#demo

Code: https://github.com/psraghothamrao8/RAD-SMART

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